

Home Automation Through Bluetooth

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Home Automation through Bluetooth' is a microcontroller-based application that can control various devices at your home using Bluetooth wireless technology. Bluetooth technology has become increasingly integrated into devices like cell phones, laptop computers and tablets, and its quick acceptance has led many to explore the possibility of various other applications that can be controlled through Bluetooth-enabled

devices. Here, the serial data transfer capability of Bluetooth technology is exploited to control devices either using your Android phone or Bluetooth-enabled laptop computer. Fig. 1 shows an overview of the complete system.

How the circuit works

Fig. 2 shows the circuit of the 'Home Automation through Bluetooth' application. The circuit is built around microcontroller P89V51RD2 (IC1), relay driver ULN2803 (IC2 through IC4), Bluetooth module HC-05, 12V and



5V voltage regulators 7812 (IC5) and 7805 (IC6), adjustable voltage regulator LM317 (IC7) and some common components.

Power supply. The circuit requires 12V, 5V and 3.3V DC supplies for various sections. 230V AC is stepped down to 15V, 1A using transformer X1, which is further rectified by the bridge rectifier BR1. The output from the bridge rectifier is filtered by capacitor C1 and then fed to the 7812 regulator to get 12V DC regulated output. The 12V output from IC5 is fed to the 5V regulator IC6, relay drivers IC2 through IC4, Bluetooth module HC-05, 12V and

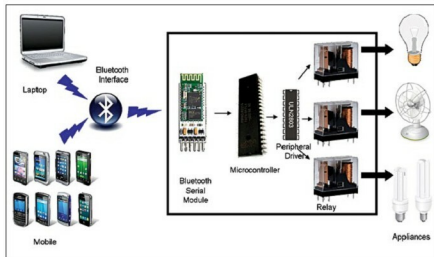


Fig. 1: Overview of the complete system

TABLE I
ASCII Values and Corresponding Actions

Character	P89V51RD2 port	Relay number	Action
A to H (upper case)	Port 0	1 to 8 (individual)	ON
a to h (lower case)	Port 0	1 to 8 (individual)	OFF
I to P (upper case)	Port 1	9 to 16 (individual)	ON
i to p (lower case)	Port 1	9 to 16 (individual)	OFF
Q to X (upper case)	Port 2	17 to 24 (individual)	ON
q to x (lower case)	Port 2	17 to 24 (individual)	OFF
Symbol '['	Port 0	All 8 relays (1 to 8 - simultaneously)	ON
Symbol ']'	Port 0	All 8 relays (1 to 8 - simultaneously)	OFF
Symbol '<'	Port 1	All 8 relays (9 to 16 - simultaneously)	ON
Symbol '>'	Port 1	All 8 relays (9 to 16 - simultaneously)	OFF
Symbol '['	Port 2	All 8 relays (17 to 24 - simultaneously)	ON
Symbol ']'	Port 2	All 8 relays (17 to 24 - simultaneously)	OFF

PARTS LIST

Semiconductors:

- IC1 - P89V51RD2 microcontroller
- IC2-IC4 - ULN2803 relay driver
- IC5 - 7812, 12V regulator
- IC6 - 7805, 5V regulator
- IC7 - LM317 adjustable voltage regulator
- LED1-LED10 - 5mm LED
- BR1 - 1A bridge rectifier module

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 680-ohm
- R2 - 390-ohm
- R3 - 220-ohm
- R4, R5 - 10-kilo-ohm
- R6-R14 - 1-kilo-ohm
- RN1-RN3 - 10-kilo-ohm resistor network

Capacitors:

- C1 - 1000µF, 35V electrolytic
- C2-C4 - 0.1µF ceramic disk
- C5 - 10µF, 16V electrolytic
- C6, C7 - 33pF ceramic disk

Miscellaneous:

- Bluetooth - HC-05 Bluetooth module
- CON1-CON3, CON7 - 8-pin connector
- CON4-CON6, CON8 - 2-pin connector
- CON9-CON17 - 2-pin 5mm terminal connector
- S1 - Tactile switch
- X1 - 230V AC primary to 15V, 1A secondary transformer
- X_{rt}, 1 - 11.0592MHz crystal oscillator